

Chapter outline

	Working mathematically				
10.01 Metric units	U	F		R	C
10.02 Precision and absolute error	U	F		R	C
10.03 Percentage error	U	F		R	C
10.04 Perimeters and areas of composite shapes	U	F	PS	R	
10.05 Areas of quadrilaterals	U	F	PS	R	
10.06 Circumferences and areas of circular shapes	U	F	PS	R	
10.07 Surface area of a prism	U	F	PS	R	
10.08 Surface area of a cylinder	U	F	PS	R	
10.09 Volumes of prisms and cylinders	U	F	PS	R	C

Wordbank

absolute error The greatest possible error of a measurement, equal to half of the precision (smallest unit) on the measuring instrument's scale

capacity The amount of fluid (liquid or gas) in a container

cross-section A 'slice' of a solid, taken across the solid rather than along it

mega- (abbreviated M) A metric prefix meaning one million

micro- (abbreviated μ) A metric prefix meaning one-millionth

precision The smallest unit on the scale of a measuring instrument

prism A solid shape with identical cross-sections and ends with straight sides

sector A region of a circle cut off by 2 radii, shaped like a slice of pizza

surface area The total area of all the faces of a solid shape

Quiz
Wordbank 10

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

CHAPTER 10 REVIEW

Language of maths

Quiz
Language of maths 10

Puzzle
Surface area and volume crossword

absolute error	arc length	base	capacity
circumference	cross-section	curved surface	diagonal
diameter	giga-	micro- (μ)	net
percentage error	perimeter	perpendicular height	pi (π)
precision	prefix	radius/radii	rectangular prism
sector	surface area	triangular prism	

1 Draw a circle and label on it:

a a sector **b** a diameter **c** an arc length

2 In the formula $V = Ah$, explain what V, A and h stand for.

3 Which metric prefix means 'one-millionth'?

4 Write the definition of a **prism** using the word **cross-section**.

5 How do you find the **surface area** of a solid?

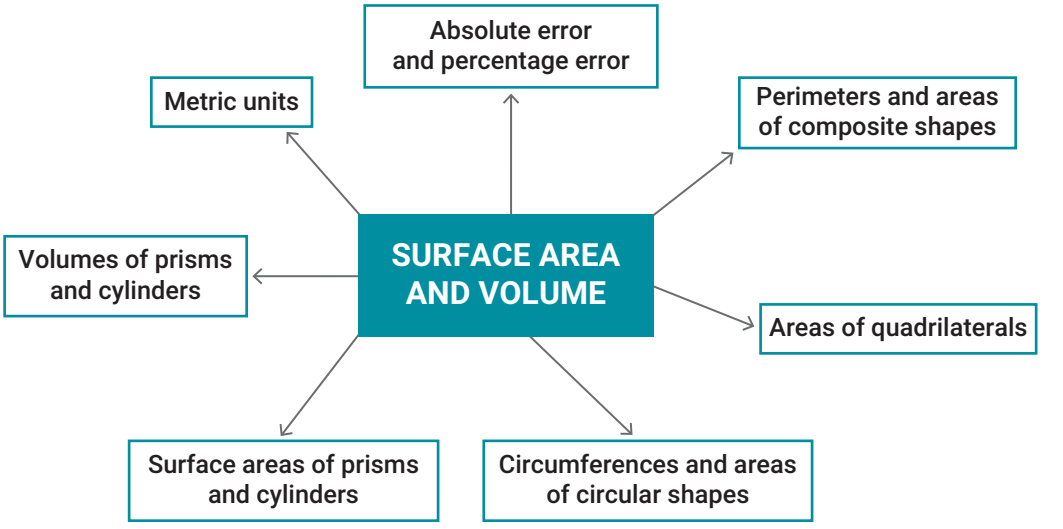
6 What is the **precision** of your ruler?

Topic summary

Worksheet
Mind map: Surface area and volume

- Write 10 questions (with solutions) that could be used in a test for this chapter. Include some questions that you have found difficult to answer.
- Swap your questions with another student and check their solutions against yours.
- List the sections of work in this chapter that you did not understand. Follow up this work with your friend or teacher.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



TEST YOURSELF 10

ANSWERS ON P. 590

1 Convert: 10.01

a 8 km to m **b** 15 min to s **c** 0.7 m to mm
d 250 000 g to kg **e** 8.4 GB to MB **f** 9.5 t to kg
g 300 min to h **h** 34 000 B to kB **i** 125 kL to L
j 2500 cm to km **k** 17 000 000 kg to t **l** 8900 GB to TB
m 10 500 ms to s **n** 9.1 millennia to years **o** 2.6 Mm to m
p 3 ly to km **q** 0.000 000 4 s to μs **r** 75 000 000 μm to m

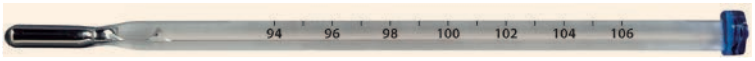
2 For each measuring instrument, state: 10.02

i its precision **ii** its absolute error.

a



b

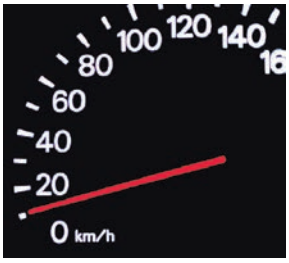


c



Shutterstock.com/Sergey Melnikov

d

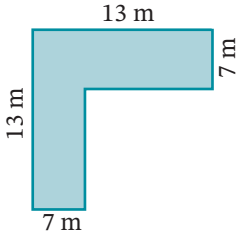


3 A super yacht is 24 m long. For this measurement, calculate: 10.03

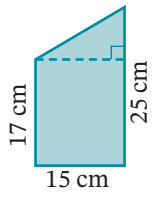
a its absolute error **b** its percentage error correct to 2 decimal places

4 Calculate the perimeter of each shape. 10.04

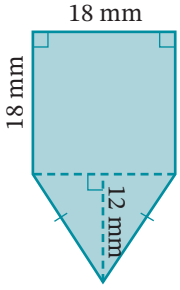
a



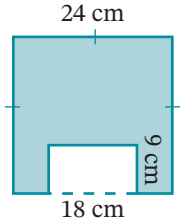
b



c



d

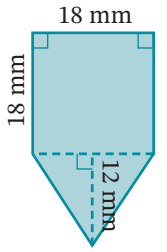


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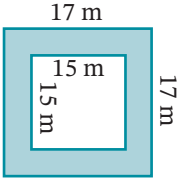
10.04

5 Calculate the area of each shape.

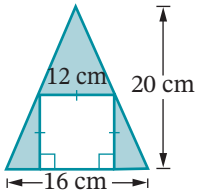
a



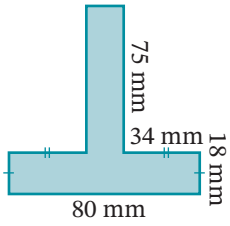
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c



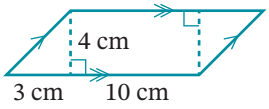
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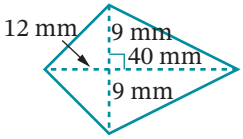
10.05

6 Calculate the area of each quadrilateral.

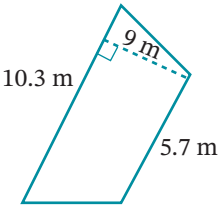
a



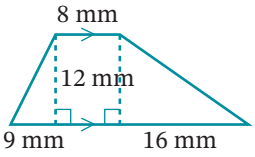
b



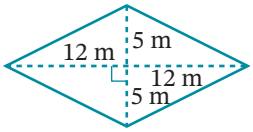
c



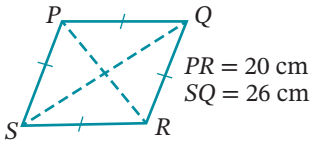
d



e



f

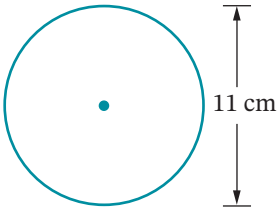


10.06

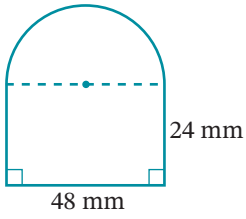
7 For each shape calculate, correct to one decimal place:

i its perimeter **ii** its area.

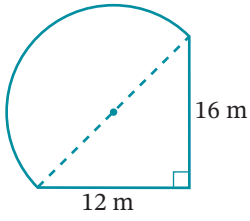
a



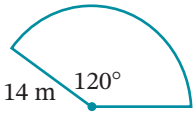
b



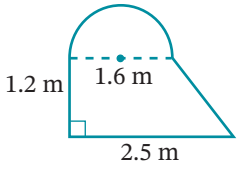
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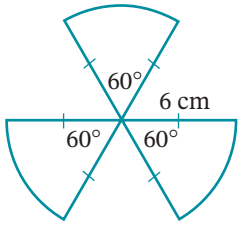
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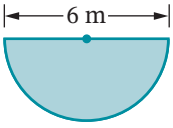
f



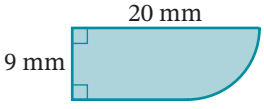
10.06

8 Calculate, correct to 2 decimal places, the area of each shape.

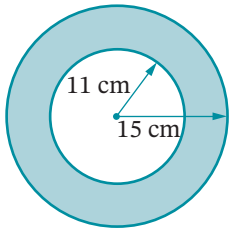
a



b



c

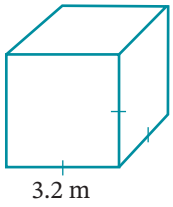


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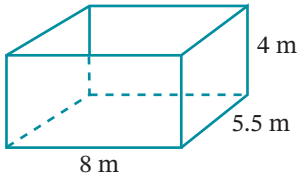
9780170479370

9 Calculate the surface area of each prism. 10.07

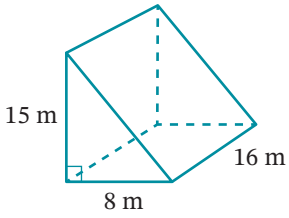
a closed cube



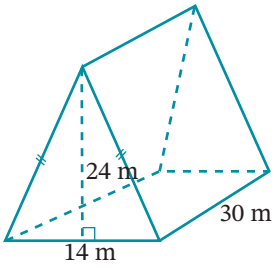
b open rectangular prism



c



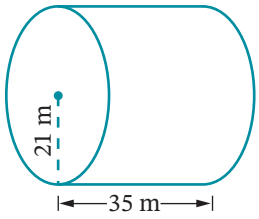
d



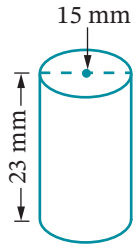
10 Calculate the volume of each prism in question **8**. 10.08

11 Calculate, correct to 2 decimal places, the surface area of each cylinder. 10.08

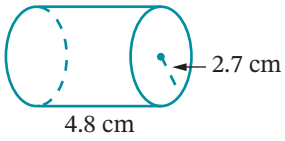
a



b



c



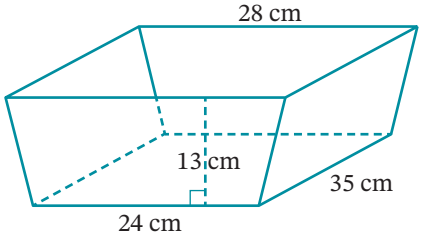
Cylinder, open at one end

12 Calculate, correct to 2 decimal places, the volume of each cylinder in question **10**. 10.09

13 A cylindrical pipe (open at both ends) is 50 cm long and has a diameter of 15 cm. Calculate its external surface area correct to the nearest square centimetre. 10.08

14 A rectangular fish tank that is 75 cm long by 55 cm wide by 35 cm deep is filled to 5 cm from the top. How many litres of water is in the tank? 10.09

15 Calculate the volume of this lunchbox. 10.09



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